JENKINS CI/CD PROJECT IMPLEMENTATION

JENKINS CI/CD PROJECT IMPLEMENTATION

Name: Harinarayana G

Course: DevOps

Date: April 2026

JENKINS CI/CD PROJECT IMPLEMENTATION

1. Project Overview

This project demonstrates a complete CI/CD (Continuous Integration and Continuous Deployment) pipeline using Jenkins integrated with multiple DevOps tools.

The pipeline automates the following processes:

- Code checkout from GitHub
- Build process using Maven
- Running unit tests
- Code quality analysis using SonarQube
- Deployment to Tomcat server
- Sending notifications via Slack
- Security scanning using Trivy

JENKINS CI/CD PROJECT IMPLEMENTATION

2. Tools Used

- Jenkins
- Apache Maven
- SonarQube
- Apache Tomcat
- Slack
- Trivy
- AWS EC2

Plugins Used

- Deploy to Container
- Pipeline Stage View
- Slack Notification
- SonarQube Scanner
- SonarQube Quality Gates

JENKINS CI/CD PROJECT IMPLEMENTATION

Architecture / Workflow

The workflow of the project is as follows:

1. Developer pushes code to GitHub repository
2. Jenkins pulls the code automatically
3. Maven builds the application
4. Unit testing is performed
5. SonarQube analyzes code quality
6. Application is deployed to Tomcat server
7. Slack sends notification to team
8. Trivy performs security scanning

JENKINS CI/CD PROJECT IMPLEMENTATION

4. Jenkins Installation

The following steps were performed to install Jenkins:

•Installed Git and Maven
•Added Jenkins repository
•Installed Java 17
•Installed Jenkins
•Started Jenkins service

Output: Jenkins server started successfully

JENKINS CI/CD PROJECT IMPLEMENTATION

SonarQube Setup

The following steps were performed:

- Downloaded SonarQube package
- Extracted the files
- Installed Java
- Created sonar user
- Started SonarQube service

Output: SonarQube dashboard accessible

JENKINS CI/CD PROJECT IMPLEMENTATION

Execution Screenshots

- Jenkins Dashboard
- Pipeline Build Success
- Console Output
- SonarQube Results
- Deployment Output
- Slack Notification
- Trivy Scan Result

Important Note

The project was successfully executed on AWS EC2.
Currently, the instance is terminated, so only screenshots are provided as proof of implementation.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;instanceState=running

aws | Search [Alt+S] | Ask Amazon Q | United States (N. Virginia) | hari (1508-5224-4953)

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- **Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager
- **Images**

Instances (1/3) Info

Last updated about 3 hours ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
JENKINS3	i-0f2c2c996f63dc92d	Running	c7i-flex.large	3/3 checks passed	View alarms +
SONAR3	i-01fa0119e80ad959d	Running	c7i-flex.large	3/3 checks passed	View alarms +
TOMCAT3	i-03446b194dec6ea40	Running	t3.micro	3/3 checks passed	View alarms +

i-01fa0119e80ad959d (SONAR3)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Not secure 44.201.143.82:8080

Jenkins

- New Item
- Build History
- Project Relationship
- Check File Fingerprint

Add description

All

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	Hari	17 min #16	N/A	20 sec
✓	☀	Project	1 min 16 sec #3	N/A	4.8 sec

Build Queue

No builds in the queue.

Build Executor Status 0/2

Icon: S M L

REST API Jenkins 2.541.3

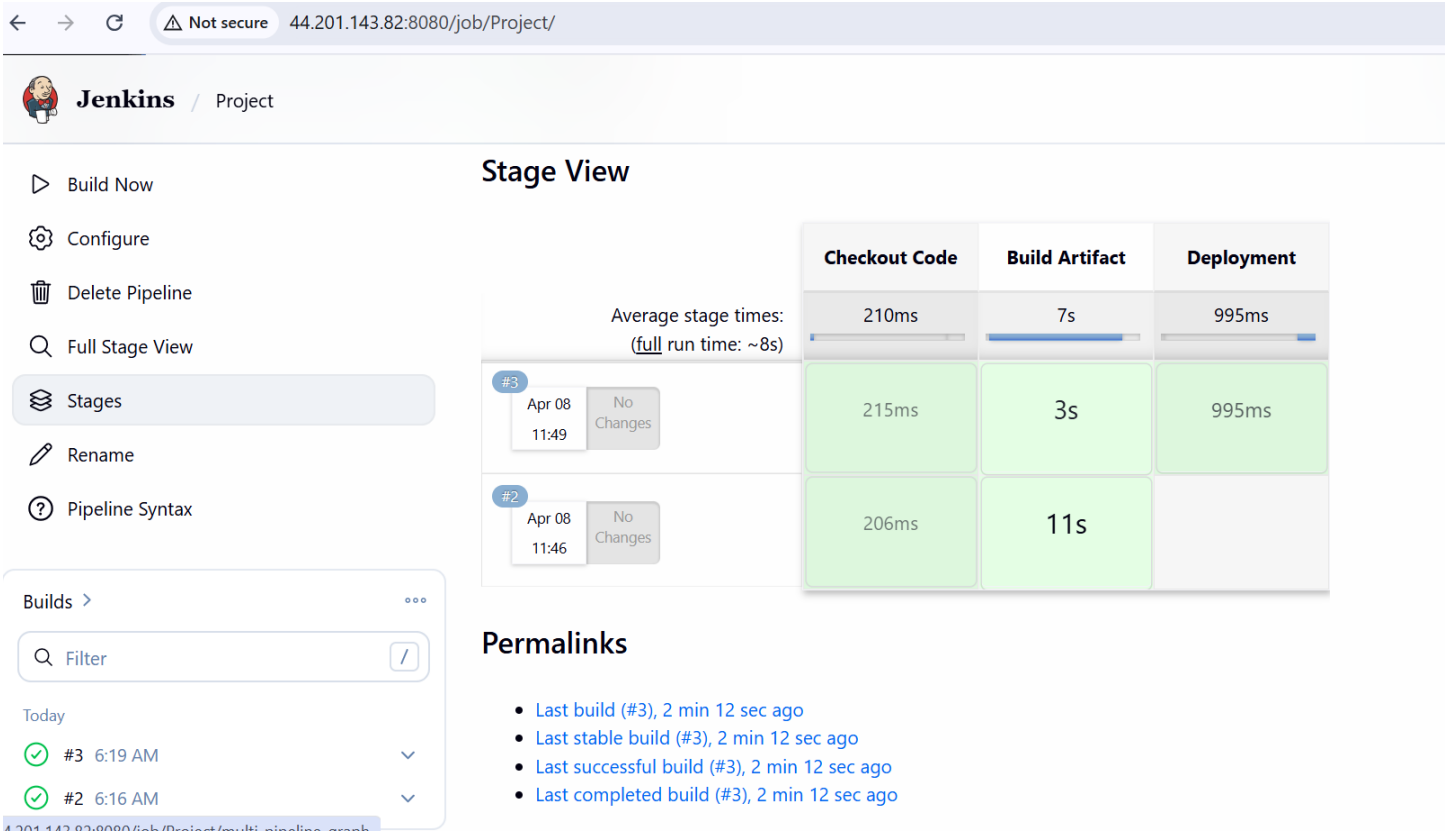
Jenkins / Hari

Today

- #16 6:03 AM
- #15 6:00 AM
- #14 5:58 AM
- #13 5:54 AM
- #12 5:46 AM
- #5 4:41 AM
- #4 4:24 AM
- #3 4:20 AM
- #2 4:12 AM

Stage View

	Declarative: Tool Install	Checkout-code	Build-project	Unit test and Coverage	Code-Analysis	Deployment	Slack-notificat
Average stage times: (full run time: ~11s)	128ms	435ms	4s	211ms	6s	272ms	202ms
#16 Apr 08 11:33 No Changes	73ms	301ms	4s	189ms	5s	267ms	209ms
#15 Apr 08 11:30 No Changes	74ms	272ms	3s	179ms	5s	243ms	198ms
#14 Apr 08 11:28 No Changes	71ms	277ms	4s	182ms	5s	260ms	191ms
#13 Apr 08 No Changes	74ms	283ms	3s	179ms	6s	305ms	210ms


Not secure
44.201.143.82:8080/job/Project/
Jenkins / Project
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#3 6:19 AM
#2 6:16 AM
Stage View
Checkout Code
Build Artifact
Deployment
Average stage times:
(full run time: ~8s)
210ms
7s
995ms
#3
Apr 08
11:49
No Changes
215ms
3s
995ms
#2
Apr 08
11:46
No Changes
206ms
11s
Permalinks
Last build (#3), 2 min 12 sec ago
Last stable build (#3), 2 min 12 sec ago
Last successful build (#3), 2 min 12 sec ago
Last completed build (#3), 2 min 12 sec ago

Not secure
44.201.143.82:8080/job/Hari/
Jenkins / Hari
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Coverage Trend
Pipeline Syntax
Builds
Filter
Hari
Add description
Test Result Trend
Passed
Skipped
Failed
1
0.5
#4
#5
#12
#13
#14
#15
#16
Code Coverage Trend
2
1
0
line covered
line missed
enlarge

app.slack.com/client/T0ARDHWRLPP/C0ARF225T0W

Describe what you are looking for

Hari

Get 55% off Slack Pro

Threads
Huddles
Directories

Starred
Drag and drop important stuff here

Channels
all-hari
jan-devops-batch
Add channels

Direct messages
Unread mentions

Home
DMs
More
Admin

jan-devops-batch

Invite teammates | Huddle

Messages | Add canvas

Today

joined jan-devops-batch.

Harinarayana G 10:46 AM
added an integration to this channel: jenkins

jenkins APP 10:49 AM
Slack/Jenkins plugin: you're all set on http://44.201.143.82:8080/

jenkins APP 11:30 AM
Hare Krishna 🙏, Project deployed successfully into Dev Env 🚀

11:33 Hare Krishna 🙏, Project deployed successfully into Dev Env 🚀

React | Reply

Message jan-devops-batch

Not secure 3.83.29.228:9000/projects

sonarqube | Projects | Issues | Rules | Quality Profiles | Quality Gates | Administration

Search for projects...

My Favorites | All

Filters

Quality Gate
Passed 1
Failed 0

Reliability (Bugs)
A 1
B 0
C 0
D 0
E 0

Security (Vulnerabilities)
A 1
B 0
C 0
D 0

Search by project name or key

Add Project

2 projects

Perspective: Overall Status

Sort by: Name

MyWebApp Passed

Last analysis: 1 hour ago

Bugs	Vulnerabilities	Hotspots Reviewed	Code Smells	Coverage	Duplications	Lines
A	A	A	A	0.0%	0.0%	76 XS XML, JSP, J...

project

Project's Main Branch is not analyzed yet. Configure analysis

2 of 2 shown

Instance details | EC2 | us-east | Console - #3 - Jenkins | Pipeline Syntax: Snippet Gene | Apache Tomcat/11.0.9 | Ask Gemini

Not secure 44.201.227.63:8080

Home | Documentation | Configuration | Examples | Wiki | Mailing Lists | Find Help

Apache Tomcat/11.0.9

APACHE SOFTWARE FOUNDATION http://www.apache.org/

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To

Server Status
Manager App
Host Manager

Developer Quick Start

Tomcat Setup
First Web Application
Realms & AAA
JDBC DataSources
Examples
Servlet Specifications
Tomcat Versions

Managing Tomcat

For security, access to the manager webapp is restricted. Users are defined in:

$CATALINA_HOME/conf/tomcat-users.xml

Documentation

Tomcat 11.0 Documentation
Tomcat 11.0 Configuration
Tomcat Wiki

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

tomcat-announce

Multi-execution mode: commands are typed to all terminals (use Ctrl+Shift+Insert to paste)

Multi-paste | Exit multi-execution mod

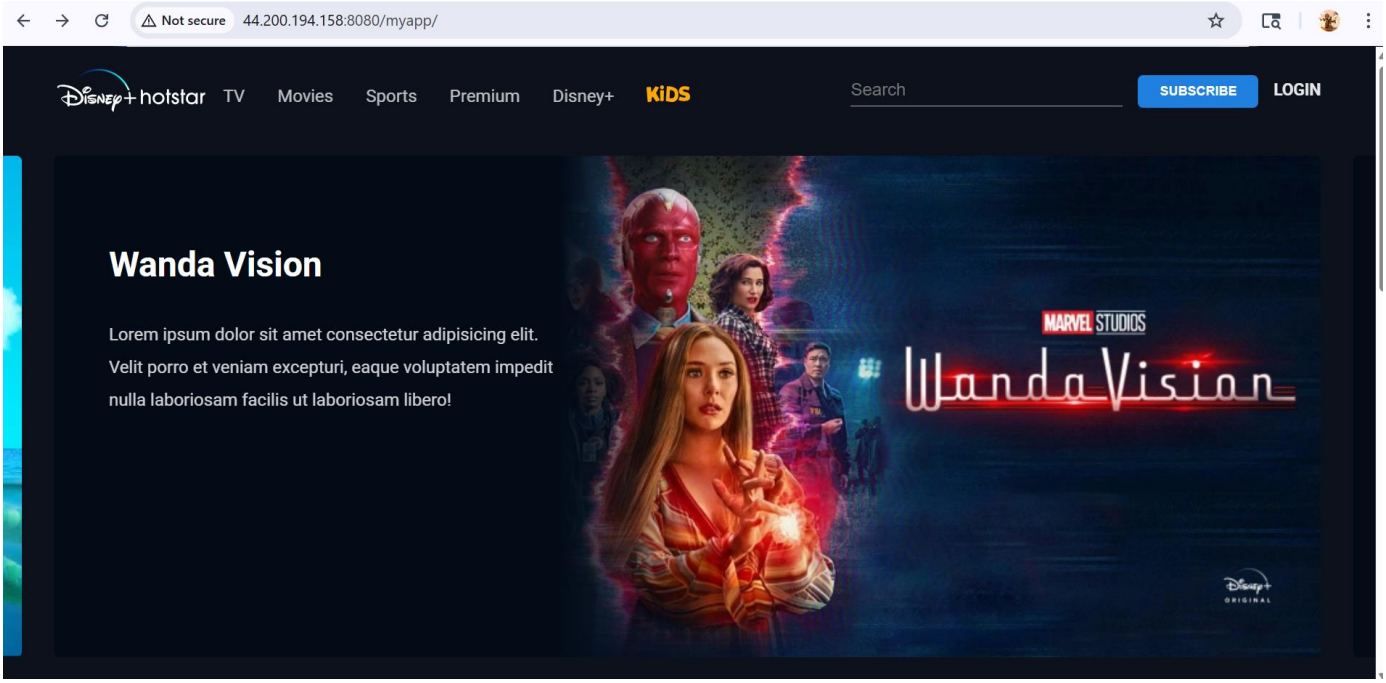
User sessions
- 3.83.29.228 (ec2-user)
- 44.200.194.158 (ec2-user)
- 44.201.143.82 (ec2-user)

```
/etc/yum.repos.d/jenkins.repo   100%[=======================

2026-04-08 04:07:33 (20.9 MB/s) - '/etc/yum.repos.d/jenkins.

Last metadata expiration check: 0:47:12 ago on Wed Apr  8 03
Package java-17-amazon-corretto-1:17.0.18+9-1.amzn2023.1.x86
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:47:12 ago on Wed Apr  8 03
Package jenkins-2.541.3-1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service
```

Exclude "JENKINS" from MultiExec mode

```
        _/m/'
Last login: Wed Apr  8 03:25:57 2026 from 1.39.180.191
[ec2-user@ip-172-31-72-36 ~]$ sudo -i
[root@ip-172-31-72-36 ~]# vi /opt/apache-tomcat-9*/conf/tomc
[root@ip-172-31-72-36 ~]# vi /opt/apache-tomcat-9*/webapps/m
[root@ip-172-31-72-36 ~]# cd /opt/apache-tomcat-9*/bin
./shutdown.sh
./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.117
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.117
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.117/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.117/bin/bootst
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/jav
AMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --a
```

```
[ec2-user@ip-172-31-91-220 ~]$ ps -ef | grep sonar
root        4354    4230  0 04:27 pts/1    00:00:00 sh sonar
root        6524    4354  0 04:28 pts/1    00:00:00 su - son
sonar       6548    6524  0 04:28 pts/1    00:00:00 -bash
sonar      28165       1  0 04:29 ?        00:00:03 /opt/son
6.50800/bin/linux-x86-64/../../conf/wrapper.conf wrapper.sys
n/linux-x86-64/./SonarQube.pid wrapper.daemonize=TRUE
sonar      28167   28165  0 04:29 ?        00:00:08 java -Ds
library.path=./lib -classpath ../../lib/sonar-application-8.
tdowner-8.9.6.50800.jar -Dwrapper.key=gc6hnpJFWB9geXXg -Dwra
ax=31999 -Dwrapper.pid=28165 -Dwrapper.version=3.2.3 -Dwrapp
meout=10 -Dwrapper.jvmid=1 org.tanukisoftware.wrapper.Wrappe
sonar      28213   28167  1 04:29 ?        00:00:38 /usr/lib
io.tmpdir=/opt/sonarqube-8.9.6.50800/temp -XX:ErrorFile=../l
workaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m
-Djna.tmpdir=/opt/sonarqube-8.9.6.50800/temp -XX:-OmitStackT
```

Exclude "SONAR" from MultiExec mode

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-lin
ux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
Last login: Wed Apr  8 06:26:41 2026 from 1.39.174.143
[ec2-user@ip-172-31-91-220 ~]$
```